FLEET TEAM DIGEST

737MAX-FTD-22-21003

Issue Title : Non-Certified Versions of Flight Control Computer Software Inadvertently Installed

Airplane Model	ATA	Minor Model(s)
737MAX	2211-00	-7, -8, -8200, -9

Other Model(s)	ECCN
	9E991

Originated Date	Last Revised Date	Created On
08/25/2021	02/24/2026	02/24/2026

Estimated Completion Date	Next Update Date
12/31/2026	09/30/2026

Revision Description

No updates were made to this Fleet Team Digest article.

Description

During the course of airplane troubleshooting, prohibited Flight Control Computer (FCC) software (SW) was inadvertently installed on a 737 MAX airplane. There are many airline operators who operate both 737NG and 737 MAX fleets. Prior to FCC SW P12.1.2, the 737NG and 737 MAX airplane models share common FCC hardware and software configurations, making it possible for a 737NG part to inadvertently become installed on a 737 MAX airplane. It is possible for older FCC software loads to be installed and flown on the 737 MAX without indication that the aircraft is in an non-certified for flight configuration.

The FCC does not currently have built-in functionality that would detect prohibited FCC software installations. Human factors, including human error or deviations from approved maintenance procedures, including the referenced Airplane Maintenance Manual task, can contribute to prohibited FCC software being installed and flown.

Background

The Enhanced Digital Flight Control System (EDFCS) provides autopilot, autothrottle, flight director, mach trim, speed trim, altitude alert, and Roll Command Alerting System (RCAS) functions. The EDFCS consists of two identical Flight Control Computers (FCCs), a Mode Control Panel (MCP), an Integrated Flight Systems Accessory Unit (IFSAU), and an optional rudder servo. The FCCs run common Operational Program Software (OPS), which is the software that provides the operating system (processor executive and power-up tests), Autopilot, Autothrottle and Speed/Mach Trim functions. At power-up each FCC reads the configuration of the Program Switch Modules (PSM), which has a set of switches which are selected to either “ON” or “OFF”, to define the appropriate aircraft model and configuration for the software to run. If the PSM configuration is determined to not be compatible with the FCC, the FCC will set the following Flight Deck Effects:

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- Autopilot Disconnect Light Illuminated
- Flight Director Flag is shown
- Speed Trim Fail Light Illuminated
- Mach Trim Fail Light Illuminated
- NO AUTOLAND message is shown (Applicable to airplanes that have fail-operational autopilot systems only)

For in-service airplanes, if a prohibited software load is installed into both FCCs, the Flight Deck Effects listed above will not be triggered. This could be mis-interpreted by maintenance and/or flight crews as the systems being operational and flight-capable.

Status

Boeing has initiated the referenced Service Related Problem (SRP) to provide a proactive enhancement for the FCC, to mitigate the potential for a prohibited FCC software to become installed onto the 737 MAX airplane. Boeing is developing changes to the 737 MAX EDFCS in conjunction with the 737-10 certification program, consisting of the following:

- Modifying the FCC Program Switch Module (PSM) DIP switch settings to add a hardware interlock.
- Modifying the FCC OPS to do a check of the hardware interlock at power-up.

Once these changes to the EDFCS become available, it will limit the possibility of the aircraft to be put into a non-conforming configuration.

Interim Action

Operators are directed to review and conform to the following AMM procedures while performing maintenance activities on the Flight Control Computer:

- AMM TASK 22-11-33-400-801, FCC Installation Procedure
- AMM TASK 22-11-33-470-801, FCC Hardware and Software Verification Procedure
- AMM TASK 22-11-33-470-802, FCC Software Installation from the Onboard Network System
- AMM TASK 22-11-33-470-802, FCC Software Installation Using the Cross-load Function

The listed AMM TASKS include detailed work instructions to ensure the correct hardware and software is installed.

Final Action

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Upon certification of the revised changes, which is currently under development in conjunction with the 737-10 certification program:

- For in-service airplanes, the referenced Boeing Service Bulletin will be made available for operators to incorporate the changes.
- For production airplanes, a production line number cut-in will be identified and the equivalent changes will be installed automatically prior to customer delivery.

Boeing understands that the FAA is considering rulemaking based on this item.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Root Cause Established: 01 July 2021 (Complete)

Solution Selected: 05 August 2021 (Complete)

Service Bulletin Available: Will follow post 737-10 ammended type certification activity completion

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Bulletin (SB)	737-22A1382		
Service Related Problem (SRP)	737MAX-SRP-22-0054		
Aircraft Maintenance Manual (AMM)	AMM TASK 22-11-33-470-801		

Related Categories

- SRPs - Safety : Service Related Problems in work by Boeing that have an airplane or personal safety impact

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